

ROBIN FEREDE

PhD Researcher

@ r.ferede@tudelft.nl

in robin-ferede

robinferede

G R. Ferede

ID 0000-0001-5158-565X



PROFILE

PhD candidate in **Aerospace Engineering** at TU Delft (MAVLab), specializing in **time-optimal quadcopter control**. My research covers **optimal control solvers** and **(model-based) reinforcement learning**, with a focus on deploying end-to-end policies on real hardware and closing the **sim-to-real gap**. Looking ahead, I want to extend this work toward **physical AI** that generalizes across tasks.

EXPERIENCE

Team Member, Control Engineer

A2RL Autonomous Drone Race

Nov 2024 – Jan 2026

Abu Dhabi, UAE

Two A2RL campaigns with the TU Delft MAVLab team:

- A2RL 2025 (*MonoRace algorithm*) – 1st AI vs Human (vs. 3 FPV champions), 1st Speed Race, 1st Drag Race, 3rd Multi-Drone Race.
- A2RL 2026 (*SkyDreamer – world-model based, pixel-to-action; technical successor to MonoRace*) – 1st Multi-Drone Race, 2nd Speed Race, 3rd AI vs Human.

Intern

Advanced Concepts Team, ESA

Sept 2020 – Dec 2020

Noordwijk, NL

Feasibility study on time optimal neural quadcopter control.

Student Team Member

Delft Aerospace Rocket Engineering

Sept 2019 – Sept 2020

Delft, NL

Working on the roll control system for the Stratos IV Rocket.

Student Team Member

Cyclone Autonomous Drone Racing

Jan 2018 – Sept 2019

Eindhoven

Developing computer vision algorithms for pose estimation.

STRENGTHS

Analytical

Teammaker

Resilient

Python

Pytorch

JAX

Stable-baselines3

C/C++

OpenCV

Robotics

AI

Reinforcement Learning

Control

Estimation

Sim-to-Real

World Models

EDUCATION

Ph.D. in Aerospace Engineering

Micro Air Vehicle Laboratory, TU Delft

Oct 2022 – now

Delft, NL

Topic: **Time Optimal Quadcopter Control**

Current research: **End-to-end RL, World Models, Sim-to-Real**

M.Sc. in Aerospace Engineering

TU Delft

2019 – 2022

Delft, NL

Internship at ESA ESTEC

Thesis: *An Adaptive Control Strategy for Neural Network based Optimal Quadcopter Controllers*, Grade: **9.5**

B.Sc. in Applied Physics

TU Eindhoven

2016 – 2019

Eindhoven, NL

Honors Academy: High Tech Systems

Thesis: *Measuring turbulence using a quadcopter with LQR control*, Grade: **8.5**

B.Sc. in Applied Mathematics

TU Eindhoven

2016 – 2019

Eindhoven, NL

Thesis: *Function approximation with deep neural networks*, Grade: **8.5**

REFEREES

Prof. Dr. Guido de Croon

@ Micro Air Vehicle Laboratory, TU Delft

✉ G.C.H.E.deCroon@tudelft.nl

Dr. Christophe De Wagter

@ Micro Air Vehicle Laboratory, TU Delft

✉ C.deWagter@tudelft.nl

Dr. Dario Izzo

@ Advanced Concepts Team, ESA

✉ Dario.Izzo@esa.int

PUBLICATIONS

Journal Articles

- D. Izzo, E. Blazquez, R. Ferede, S. Origer, C. De Wagter, and G. C. de Croon, "Optimality principles in spacecraft neural guidance and control," *Science Robotics*, vol. 9, no. 91, eadi6421, 2024. [Online]. Available: <https://doi.org/10.1126/scirobotics.adi6421>.
- R. Ferede, G. de Croon, C. De Wagter, and D. Izzo, "End-to-end neural network based optimal quadcopter control," *Robotics and Autonomous Systems*, p. 104 588, 2023. [Online]. Available: <https://doi.org/10.1016/j.robot.2023.104588>.

Conference Proceedings

- M. van Breukelen Castillo, R. Ferede, R. Vos, C. De Wagter, and G. C. de Croon, "Spiking neural networks for high-speed continuous quadcopter control using proximal policy optimization," in *16th International Micro Air Vehicle Conference and Competition (IMAV)*, Best Paper Award, 2025. [Online]. Available: <https://www.imavs.org/papers/2025/17.pdf>.
- R. Ferede, T. Blaha, E. Lucassen, C. De Wagter, and G. C. de Croon, "One net to rule them all: Domain randomization in quadcopter racing across different platforms," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [Online]. Available: <https://arxiv.org/abs/2504.21586>.
- R. Ferede, C. De Wagter, D. Izzo, and G. C. De Croon, "End-to-end reinforcement learning for time-optimal quadcopter flight," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2024, pp. 6172–6177. [Online]. Available: <https://doi.org/10.1109/ICRA57147.2024.10611665>.

Preprints

- S. Bahnam, R. Ferede, T. Blaha, *et al.*, *Monorace: Winning champion-level drone racing with robust monocular ai*, Under review at *Science Robotics*, 2025. [Online]. Available: <https://arxiv.org/abs/2601.15222>.
- A. Verraest, S. Bahnam, R. Ferede, G. C. de Croon, and C. De Wagter, *Skydreamer: Interpretable end-to-end vision-based drone racing with model-based reinforcement learning*, 2025. [Online]. Available: <https://arxiv.org/abs/2510.14783>.
- S. Origer, C. De Wagter, R. Ferede, G. C. de Croon, and D. Izzo, *Guidance & control networks for time-optimal quadcopter flight*, 2023. [Online]. Available: <https://arxiv.org/pdf/2305.02705>.